

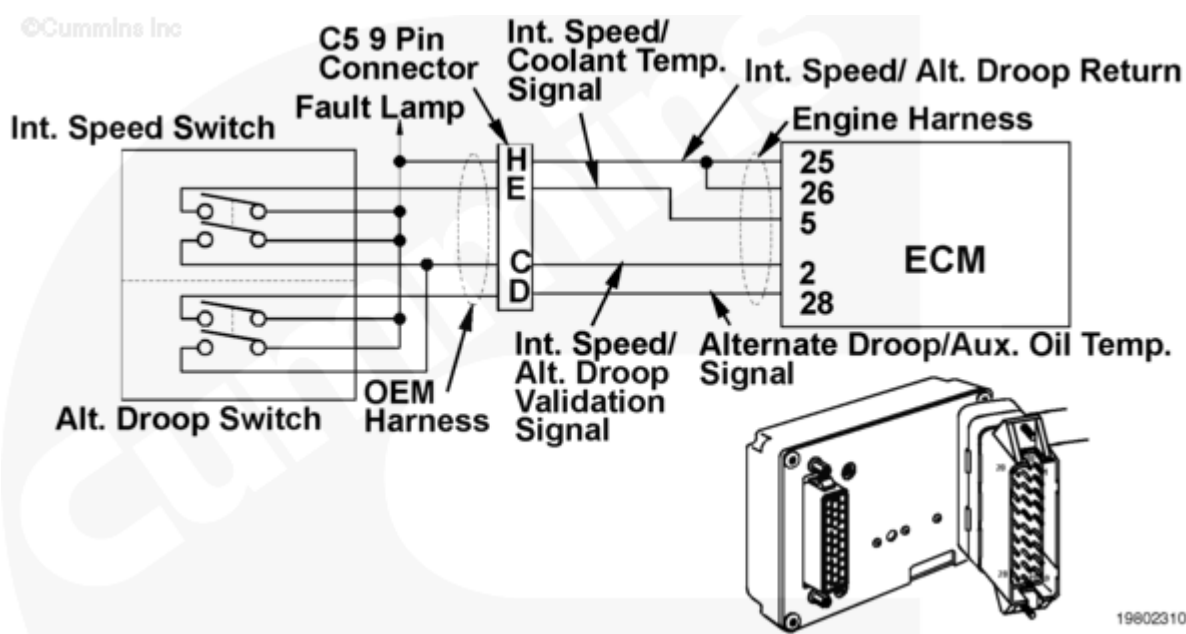
Fault Code: 445

Intermediate-Speed/Alternate Droop Validation Signal Circuit - Voltage Below Normal or Shorted to Low Source

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Overview

Codes	Reason	Effect
Fault Code: 445 PID(P): S254 SPN: FMI: 3 Lamp: On SRT: 00-639	The intermediate-speed/alternate droop validation signal is on and the intermediate-speed signal or alternate droop signal is off.	Loss of intermediate-speed and alternate droop features.



Intermediate-Speed/Alternate Droop Validation Signal Circuit

Circuit Description

The intermediate-speed/alternate droop validation signal is a safety feature that enables the intermediate-speed and alternate droop features while the validation signal is detected as a high voltage. The intermediate-speed switch and alternate droop switch provide the intermediate-speed, alternate droop, and intermediate-speed/alternate droop validation signals to the ECM through the OEM harness and the main engine harness.

Component Location

The intermediate-speed switch and the alternate droop switch location may vary and is OEM dependent.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected low voltage at the intermediate-speed/alternate droop validation signal, and high voltage detected at the intermediate-speed signal or high voltage detected at the alternate droop signal.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE light when the diagnostic runs and fails.
- No engine protection for engine oil pressure.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.
- No engine protection for engine engine oil pressure.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

- This fault code will **not** be active while the intermediate-speed or alternate droop switches are in the OFF position.

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Malfunctioning or damaged OEM wiring harness.
- Damaged or loose connectors.
- Malfunctioning or damaged alternate torque idle switch.
- Malfunctioning or damaged intermediate-speed switch (retard).
- Malfunctioning or damaged alternate droop switch.

Refer to Troubleshooting Fault Code t05-445 (</qs3/pubsys2/xml/en/procedures/98/98-t05-445.html>)

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